

Notes for ‘Odd and even functions’

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**Important Ideas and Useful Facts:**

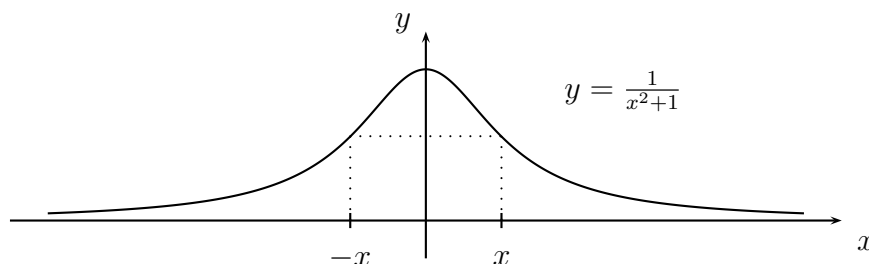
- (i) **Even functions:** An *even function* has the property that its graph has *reflectional symmetry about the y-axis*, by which we mean that if we reflect the curve in the y-axis then we get the curve back again. If  $y = f(x)$  is a function, then this is equivalent to the algebraic condition that if  $x$  is in the domain of  $f$  then  $-x$  is also in the domain and

$$f(-x) = f(x) .$$

For example, if  $f(x) = \frac{1}{x^2 + 1}$ , the graph is the familiar *witch of Maria Agnesi*, and

$$f(-x) = \frac{1}{(-x)^2 + 1} = \frac{1}{x^2 + 1} = f(x) ,$$

so the function is even.



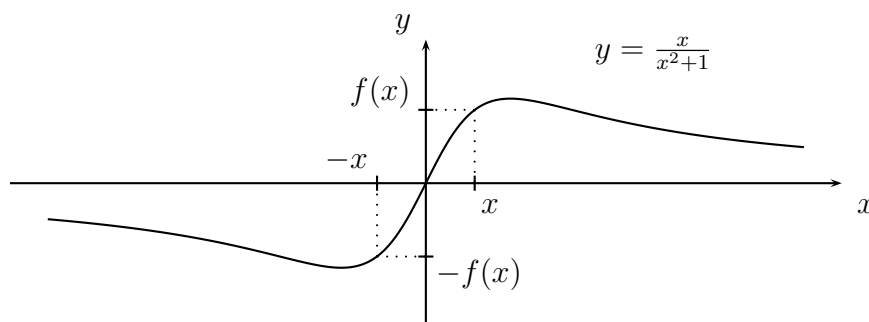
- (ii) **Odd functions:** An *odd function* has the property that its graph has  $180^\circ$  *rotational symmetry about the origin*, by which we mean that if we rotate the curve  $180^\circ$  about the origin then we get the curve back again. If  $y = f(x)$  is a function, then this is equivalent to the algebraic condition that if  $x$  is in the domain of  $f$  then  $-x$  is also in the domain and

$$f(-x) = -f(x) .$$

For example, if  $f(x) = \frac{x}{x^2 + 1}$ , the graph is *Newton's serpentine*, and

$$f(-x) = \frac{-x}{(-x)^2 + 1} = -\frac{x}{x^2 + 1} = -f(x) ,$$

so the function is odd.



- (iii) **Integrating an even function over a symmetric interval:** An interval in the real line is called *symmetric* if it has the form  $[-a, a]$  for some real number  $a$ . The area under the curve of an even function  $y = f(x)$  over a symmetric interval  $[-a, a]$  is twice the area over the nonnegative interval  $[0, a]$ , that is,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx .$$

For example,

$$\int_{-1}^1 \frac{dx}{x^2 + 1} = 2 \int_0^1 \frac{dx}{x^2 + 1} = 2 [\tan^{-1} x]_0^1 = 2 \times \left( \frac{\pi}{4} - 0 \right) = \frac{\pi}{2} .$$

- (iv) **Integrating an odd function over a symmetric interval:** The area under the curve of an odd function  $y = f(x)$  over a symmetric interval  $[-a, a]$  is zero, that is,

$$\int_{-a}^a f(x) dx = 0 .$$

For example, we get the following automatically:

$$\int_{-1}^1 \frac{x}{x^2 + 1} dx = 0 ,$$

without any need to antidifferentiate the integrand.

- (v) **Arithmetic of odd and even functions:** The sum, difference, product or quotient of two even functions is even.

For example,  $y = x^2$  and  $y = \cos x$  are both even, so the following are even also:

$$y = x^2 + \cos x , \quad y = x^2 - \cos x , \quad y = x^2 \cos x , \quad y = \frac{x^2}{\cos x} , \quad y = \frac{\cos x}{x^2} .$$

The sum and difference of two odd functions is odd, but the product and quotient of two odd functions is even.

For example,  $y = \sin x$  and  $y = x$  are both odd, so the following are odd also:

$$y = x + \sin x , \quad y = x - \sin x , \quad -x + \sin x ,$$

but the following are even:

$$y = x \sin x , \quad y = \frac{x}{\sin x} , \quad y = \frac{\sin x}{x} .$$

The product or quotient of an even function with an odd function is odd.

For example,  $y = \cos x$  is even and  $y = \sin x$  is odd, so the following are odd:

$$y = \cos x \sin x , \quad y = \frac{\sin x}{\cos x} = \tan x , \quad y = \frac{\cos x}{\sin x} .$$

The inverse  $f^{-1}$  of an invertible odd function  $f$  is odd. This follows because if  $x$  is in the domain of  $f^{-1}$  then  $x = f(a)$  for some  $a$  so that

$$-x = -f(a) = f(-a) ,$$

since  $f$  is odd, so that  $-x$  is in the domain of  $f^{-1}$ , and further

$$f^{-1}(-x) = -a = -f^{-1}(x) .$$

For example, the function  $y = \sin x$  is odd and invertible when restricted to the domain  $[-\frac{\pi}{2}, \frac{\pi}{2}]$ , so that the inverse function  $y = \sin^{-1}(x)$  is also odd.

## Examples and derivations:

1. Taking the sum or difference of an even function with an odd function may produce a function that is neither odd nor even. For example,  $y = x$  is odd and  $y = 1 + x^2$  is even. Let  $f$  and  $g$  be functions with the rules

$$f(x) = 1 + x + x^2 \quad \text{and} \quad g(x) = 1 - x + x^2 .$$

Then  $f$  is neither even nor odd. For example,  $f(1) = 3$ , whilst  $f(-1) = 1$ , which does not equal  $f(1)$  or  $-f(1)$ . Similarly,  $g$  is neither even nor odd. For example,  $g(1) = 1$ , whilst  $g(-1) = 3$ , which does not equal  $g(1)$  or  $-g(1)$ .

2. Let  $f$  be the function with rule  $y = f(x) = x^n$ , where  $n$  is an integer. If  $n$  is even, say  $n = 2k$ , for some integer  $k$ , then

$$f(-x) = (-x)^n = (-1)^n x^n = (-1)^{2k} x^n = ((-1)^2)^k x^n = (1)x^n = x^n = f(x) ,$$

which shows that  $f$  is even. For example, the following functions are even:

$$y = 1 = x^0, \quad y = x^8, \quad y = x^{-2} = \frac{1}{x^2}, \quad y = x^{-8} = \frac{1}{x^8} .$$

On the other hand, if  $n$  is odd, say  $n = 2k + 1$ , for some integer  $k$ , then

$$f(-x) = (-x)^n = (-x)^{2k+1} = (-x)^{2k}(-x) = x^{2k}(-x) = -x^{2k+1} = -f(x) ,$$

which shows that  $f$  is odd. For example, the following functions are odd:

$$y = x = x^1, \quad y = x^9, \quad y = x^{-3} = \frac{1}{x^3}, \quad y = x^{-9} = \frac{1}{x^9} .$$

3. From the previous example, and the arithmetic of even and odd functions, it follows that any polynomial or rational function involving only even powers of  $x$  is even. For example, the following are even:

$$y = 1 + 2x^2 - 3x^4 + 7x^6 \quad \text{and} \quad y = \frac{3 - 7x^2 + 5x^4 - x^{10}}{1 - x^2 + x^6 - x^{12}} .$$

Note that each of the constants (even if odd integers) represent even functions!

Any polynomial involving only odd powers of  $x$  will be odd. For example, the following function is odd:

$$y = 2x + 3x^5 - 6x^7 + x^{99} .$$

The situation involving mixtures of odd and even, when forming rational functions, is more delicate. If  $y = f(x) = \frac{p(x)}{q(x)}$  is a rational function, say, a ratio of polynomials  $p(x)$  and  $q(x)$ , then  $f$  is even if both  $p(x)$  and  $q(x)$  are odd, but odd if only one of  $p(x)$  and  $q(x)$  is even and the other is odd. For example, the following function is even:

$$y = \frac{x - x^3 + x^7}{x + x^5 + x^9} ,$$

and the following functions are odd:

$$y = \frac{x - x^3 + x^7}{1 + x^4 + x^8} \quad \text{and} \quad \frac{1 + x^4 + x^8}{x - x^3 + x^7} .$$

4. Verify algebraically that if  $f$  and  $g$  are odd with the same domain, then  $h$  is even, where  $h$  has the rule  $h(x) = f(x)g(x)$ .

*Solution:* For  $x$  in the domain of  $h$ , we have, using the fact that both  $f$  and  $g$  are odd,

$$h(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = h(x).$$

5. Verify algebraically that if  $f$  is even and  $g$  is odd with the same domain, then  $h$  is odd, where  $h$  has the rule  $h(x) = f(x)g(x)$ .

*Solution:* For  $x$  in the domain of  $h$ , we have, using the fact that  $f$  is even and  $g$  is odd,

$$h(-x) = f(-x)g(-x) = f(x)(-g(x)) = -f(x)g(x) = -h(x).$$

6. This exercise is very difficult and challenging, and included to demonstrate the surprising power of the notions of even and odd functions. Find

$$\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1 + e^{-x}} dx.$$

As a hint, if you graph the curve  $y = \frac{1}{1+e^{-x}}$ , you will discover that it has  $180^\circ$  rotational symmetry about the point of inflection, so can be turned into an odd function by an appropriate translation in the plane.

*Solution:* We begin by sketching the curve

$$y = \frac{1}{1 + e^{-x}} = (1 + e^{-x})^{-1}.$$

By the Chain Rule,

$$y' = -(1 + e^{-x})^{-2}(-e^{-x}) = \frac{e^{-x}}{(1 + e^{-x})^2}.$$

which is clearly positive for all  $x$ . Hence the curve is increasing over all of the real line  $\mathbb{R}$ . By the Quotient Rule,

$$\begin{aligned} y'' &= \frac{(1 + e^{-x})^2(-e^{-x}) - (e^{-x})(2)(1 + e^{-x})(-e^{-x})}{(1 + e^{-x})^4} \\ &= \frac{-e^{-x}(1 + e^{-x}) + 2(e^{-x})(e^{-x})}{(1 + e^{-x})^3} = \frac{e^{-x}(e^{-x} - 1)}{(1 + e^{-x})^3} = \frac{e^{-2x}(1 - e^x)}{(1 + e^{-x})^3}, \end{aligned}$$

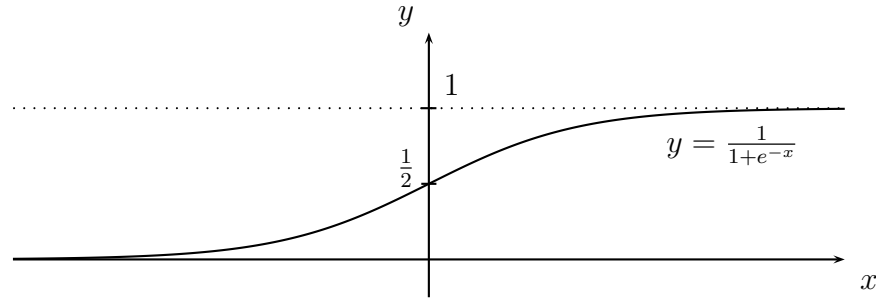
which is zero only when  $1 - e^x = 0$ , that is, when  $e^x = 1$ , giving  $x = 0$ . Further, the sign of  $y''$  is the same as the sign of  $1 - e^x$ , so we get the following sign diagram:

$$\begin{array}{c|ccc} x & & 0 & \\ \hline y'' & & 0 & \\ & + & & - \\ & \cup & & \cap \end{array}$$

Thus the curve is increasing with an inflection at  $x = 0$ , that is at the  $y$ -intercept  $y(0) = \frac{1}{2}$ . Observe that

$$\lim_{x \rightarrow \infty} \frac{1}{1 + e^{-x}} = 1 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{1 + e^{-x}} = 0,$$

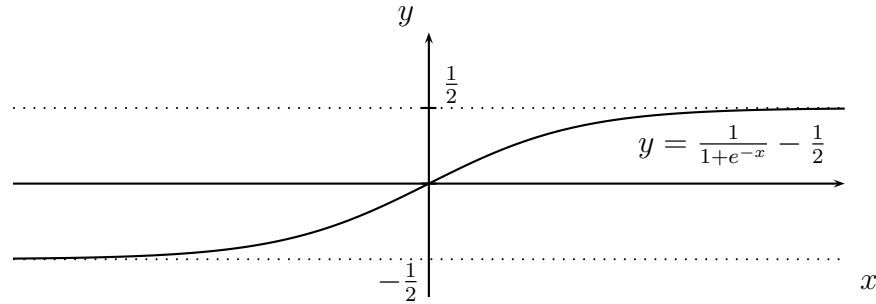
since  $e^{-x} \rightarrow 0$  as  $x \rightarrow \infty$ , and  $e^{-x} \rightarrow \infty$  as  $x \rightarrow -\infty$ . Thus the  $x$ -axis and the line  $y = 1$  are horizontal asymptotes, and we get the following sketch:



The graph appears to have 180° rotational symmetry about the point of inflection, so if we shift the graph uniformly  $\frac{1}{2}$  vertically downwards, we should be able to get rotational symmetry about the origin. We therefore form the function  $f$  defined by the rule

$$f(x) = \frac{1}{1+e^{-x}} - \frac{1}{2} = \frac{2 - (1+e^{-x})}{2(1+e^{-x})} = \frac{1-e^{-x}}{2(1+e^{-x})},$$

whose graph should look as follows:



If we are right, then  $f$  should be an odd function. How can we check? We observe that, for any input  $x$ ,

$$f(-x) = \frac{1 - e^{-(-x)}}{2(1 + e^{-(-x)})} = \frac{1 - e^x}{2(1 + e^x)} = \frac{e^{-x} - 1}{2(e^{-x} + 1)} = -\frac{1 - e^{-x}}{2(1 + e^{-x})} = -f(x),$$

which proves, without any doubt, that  $f$  is odd. Hence the function  $y = f(x) \cos x$  must be odd, since  $y = \cos x$  is even. Now we can return to the problem at hand and calculate as follows, exploiting the fact that the definite integral of an odd function over a symmetric interval is zero, and we double the definite integral of an even function over the nonnegative half of the interval:

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^{-x}} dx &= \int_{-\pi/2}^{\pi/2} \cos x \left( \frac{1}{1+e^{-x}} \right) dx = \int_{-\pi/2}^{\pi/2} \cos x \left( \frac{1}{1+e^{-x}} - \frac{1}{2} + \frac{1}{2} \right) dx \\ &= \int_{-\pi/2}^{\pi/2} \cos x \left( f(x) + \frac{1}{2} \right) dx = \int_{-\pi/2}^{\pi/2} f(x) \cos x dx + \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos x dx \\ &= 0 + \frac{2}{2} \int_0^{\pi/2} \cos x dx = [\sin x]_0^{\pi/2} = \sin\left(\frac{\pi}{2}\right) - \sin 0 = 1. \end{aligned}$$